A look back

Let us take a step back and see what we have been doing so far.

We started off with impulse response of a system. Recall that impulse response of a system is the output of a system due to an impulse. For discrete-time systems impulse is nothing but a unit sample. Such samples are practically possible. But for continuous systems an impulse is described as follows —

$$\left.\begin{aligned} \delta(t) &= 0 \quad \text{for } t \neq 0 \\ \text{and} \quad \int_{-\infty}^{\infty} \delta(t)\,dt &= 1 \end{aligned}\right\}$$

The presence of such ~~ss~~ signals seem practically impossible. However, it is more of a "relative" thing. An impulse is any signal that is entirely zero except for a short blip of arbitrary shape. For example, an impulse to a microwave transmitter may have to be in the picosecond range because the electronics responds in nanoseconds. In comparison, a volcano that erupts after years may be a perfectly good impulse to geological changes that take millenia.

But the question is why did we study impulse response? This is because impulse response completely describes a system. If we know the impulse response of a system we can find the output of a LTI system "easily" —

$$y(t) = \underset{\text{input}}{x(t)} * \underset{\text{impulse response}}{h(t)}$$

To understand impulses let us look into the night sky at a planet and a star. Both appear about the same brightness and size to the unaided eye. But this should not be the case right. Because a planet, (say Mars), is about 6000 km in diameter and is 60 million kilometers from earth. In comparison, a star, say Sirius is about 300 times larger and over one-million times farther away. Should not Mars appear much bigger than Sirius? This is not the case because the planet and star is small enough to be impulses to the human visual system. The perceived shape is the impulse response of the eye, not the actual image

of the star or planet. This becomes obvious when the two objects are viewed through a small telescope; Mars appear as a dim disk while Sirius still appears as a bright impulse.

We have also looked into Fourier transform. Fourier transform shows that "almost all" signals can be represented as sum of sinusoids. Since each sinusoid corresponds to a unique frequency, FT maps a time-domain signal to its frequency-domain counterpart. But why do we have to do so? And if so, why sinusoids - why not as sum of other waves like square or triangular?

The various methods that we have studied are ways to simplify analysis of the interaction of a system with a signal. Fourier transform is also such a method. For example, previously we have written signals as sum/integral of impulses -- that helped us to get the idea of convolution.

Let us do a thought experiment. Suppose two persons A and B are playing a piano and both of them presses two different keys. If you could see sound, it would look like air molecules

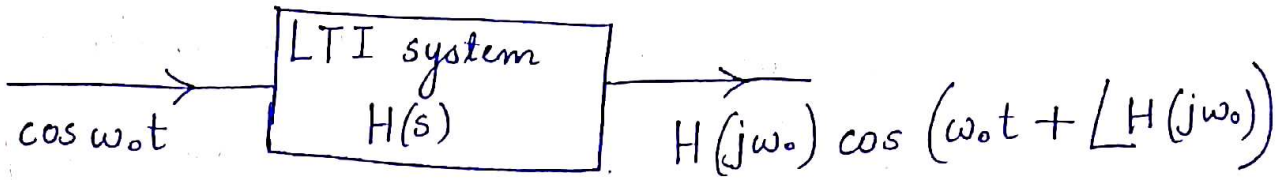
bouncing back and forth very quickly. But oddly enough, when you heard the two sounds you hear two different sounds. What differentiates both these sounds is the frequency? But how do we mathematically show this. We use Fourier transform.

But why sinusoids? There are infinite number of ways that a signal can be decomposed. The goal of decomposition is to end up with something easier to deal with than the original signal. For example, impulse decomposition allows the signal to be examined one point at a time. The component sine and cosine waves are simpler than the original signal because they have a property which the original signal does not have: sinusoidal fidelity. What is this —

a sinusoidal input to a LTI system produces a sinusoidal output of the same frequency but with different amplitude and phase.

— This is due to the eigenfunction property

of complex exponentials — we talked about this in details in module - 7.



So, we can easily compute the output. Here is another example (there are many of course) —

Have you ever noticed that each of your phone's number buttons sounds different when you are pressing them during a call (say while talking to an IVR) and it sounds the same for every phone model? That is because they are composed of two different sinusoids which can be used to uniquely identify the button. When you use use your phone to punch in combinations to navigate a menu, the way that the other side (or IVR) knows what keys you pressed is by doing a Fourier transform of the input and looking at the frequencies present. This is known as DTMF - Dual tone multi frequency technique.

Then why did we have to study Laplace transform and z-transform.

Such transforms give us a method to analyze the system. Using such transforms, we can easily comment on the stability and other properties of the system. Recall that the positions of the poles divulge a lot about the systems — see module 3 and module 4. Further, these tools can also be used to solve differential equations and difference equation using polynomials.

Notice that for a signal $x(t)$ the Fourier transform is —

$$X(j\omega) = \int_{-\infty}^{\infty} x(t) e^{-j\omega t} dt \longrightarrow (1)$$

The Laplace transform is

$$X(s) = \int_{-\infty}^{\infty} x(t) e^{-st} dt \longrightarrow (2)$$

Recall that s is any complex number. So, we can write $s = \alpha + j\omega$. Putting this in equation (2),

$$X(s) = \int_{-\infty}^{\infty} x(t) e^{-(\alpha + j\omega)t} dt$$

$$= \int_{-\infty}^{\infty} \left(x(t) e^{-\alpha t}\right) e^{-j\omega t} dt \longrightarrow (3)$$

Comparing (3) & (1), we can conclude that the Laplace transform of a signal $x(t)$ is nothing but the Fourier transform of $x(t) e^{-\alpha t}$.

Remember Fourier transform is generally used to understand the qualities of a signal (e.g. audio signal, image etc.) and Laplace transform is used to analyze systems. Mathematically, Fourier transform seems to be a subset of Laplace transform because FT is just LT at $\alpha = 0$ in equation (3). We will use all this tools extensively after this module.

Another intuition about time-domain and frequency domain is —

* High frequencies come in where the signal changes rapidly. At the extreme, when a signal changes suddenly, infinitely high frequencies come into play.

* Low frequencies come in where the signal changes slowly. At the extreme, when it doesnot change at all, we have DC component (zero frequency).

So, if you can make a signal smoother and smoother it will have lower frequencies. Remember the square wave, it was discontinuous and we had frequency components till infinity. Noise have high frequency components, since they have rapid transitions.

From here onwards we will look into processing of signals. Till now we have been looking into properties of signals and systems. In the process developed several tools - we use this next in signal processing.

Now that you know Fourier transform and series.
Here is some thing to laugh about.

Fourier's theorem can also be put to nonscientific uses. Suppose one takes the outline of a human face and forms a periodic graph by repeating the profile (fig. 109). Fourier's theorem says that the formula of this periodic function is a sum of simple sine functions, each with a definite period and amplitude. Each of these sine functions can be matched by a simple sound and the combination of these sounds will yield a musical sound. Hence there is a musical sound determined by the repeated facial profile. The young man who tells his girl that her face reminds him of a melody is telling the truth—if he knows Fourier's theorem. Herein lies the real charm of Fourier's theorem.

Mathematics and the Physical World - Morris Kline